Reddy Hema Sri Asritha

AI & Data Science Graduate | GenAI | LLM Evaluation | RAG | Python | SQL

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Summary

AI & Data Science graduate specializing in **Generative AI, RAG systems, and LLM evaluation**. Built RAG pipelines with **LangChain, LangGraph, FAISS, and Groq**. Published **4 IEEE papers** and **1 SSRN preprint**. Achieved **PSNR >65 dB and SSIM ≈ 1.0**, exceeding evaluated baseline methods. Proficient in Python, NLP, SQL, Power BI, and AI model evaluation.

Education

VR Siddhartha Engineering College **2022 – 2026**

B.Tech in Artificial Intelligence and Data Science (CGPA: 7.55) *Andhra Pradesh, India*

Experience

Innodata India Pvt. Ltd. **June 2026 – Aug 2026**

Analyst – AI/LLM Practice *Remote*

- Worked across **large-scale multimodal AI datasets** spanning **text, audio, image, and video** modalities, applying **task-specific annotation guidelines and evaluation criteria** for each data type.
- Performed **LLM data annotation, labeling, and quality assurance** maintaining **high accuracy and consistency** across multiple concurrent AI training pipelines.
- Adapted to **different quality rubrics, taxonomies, and evaluation frameworks** per project to support **AI model training, alignment, and quality improvement** workflows.

UV Netware Private Limited **Mar 2026 – Apr 2026**

Database Analyst Intern *Remote*

- Executed **SQL-based analytical queries** on **1,000+** enterprise records; built **data validation and cleaning pipelines** ensuring governance compliance.

Consilience Research Foundation **July 2025 – Sep 2025**

Research Intern *Remote*

- Applied **NLP (TF-IDF, LDA)** to analyze **50+ policy documents**; built PM2.5 vulnerability index using WHO thresholds and Python data pipelines.

Projects

Company Intelligence Agent | *LangGraph, LangChain, Groq, FAISS, RAG* **GitHub**

- Built RAG pipeline with **NVIDIA embeddings**, FAISS retrieval, and metadata filtering; used **LangGraph + Groq** for multi-tool orchestration and reasoning.

CNN-Based Laplacian Residual Network for Image Steganalysis | *PyTorch, CNN* **IEEE 2026 | GitHub**

- Developed CNN-based steganalysis model using Laplacian residual features for detecting hidden data in digital images. Published IEEE ICCVBIC 2026.

CTP-RDH: Context–Target Prediction based Reversible Data Hiding | *CNN, PyTorch* **IEEE 2026 | GitHub**

- Designed adaptive CNN for reversible data hiding using context-target prediction; achieved near-lossless image recovery. Published IEEE ICSSIT 2026.

CTP-RDHNet: Enhanced Interpolation-Based RDH | *PyTorch, CNN, OpenCV* **IEEE 2026 | GitHub**

- Achieved **PSNR > 65 dB** and **SSIM ≈ 1.0**, with **12%+** improvement over **5+ baselines**. Published IEEE ICVADV 2026.

RFID-Driven Smart Shopping Trolley | *Arduino, RFID, IoT* **IEEE 2024 | GitHub**

- Developed RFID-based automated billing using Arduino, reducing checkout time by **60%**. Published IEEE ICCES 2024.

Technical Skills

Languages: Python, SQL, Java, C

AI & LLM: Generative AI, LLM Evaluation, RAG, LangChain, LangGraph, Prompt Engineering, Data Annotation, Multimodal AI, Responsible AI

RAG & Retrieval: FAISS, Vector Databases, Embeddings, Semantic Search, Document Chunking, NVIDIA Embeddings, Groq API

ML & DL: PyTorch, TensorFlow, scikit-learn, OpenCV, CNNs, NLP (TF-IDF, LDA), Feature Engineering

Data & BI: SQL, Power BI, DAX, Power Query, Excel, Pandas, NumPy, Matplotlib

Tools & Cloud: Git, GitHub, Jupyter, LaTeX, AWS, Google Cloud

Achievements & Certifications

AIR 1547 – TCS CodeVita (100,000+ participants) | **MSME 4.0 Hackathon** – Selected Nationally | **Theme Lead** – National AI Hackathon 2025

5 Publications: IEEE'26 Steganalysis | IEEE'26 CTP-RDH | IEEE'26 RDH | IEEE'24 RFID | SSRN'26 PM2.5